

Homework I - Task 26: Hypernym Discovery

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1 Introduction

Hypernym Discovery [1] is an NLP classification task that focuses on retrieving a list of the most appropriate hypernyms from a text corpus given a term as input. Specifically, given a word w having y , z and u as hypernyms, the aim of the work is to create one **entry** for each hypernym by adding three **distractors** $d1$, $d2$ and $d3$. The term "distractors" refers to words that have *similar contextual meaning* as the hypernyms and they are used to introduce confusion within the model and test its ability to detect right hypernyms.

2 Dataset Format

From the provided EVALITA dataset, only the training and test folders have been considered. Within each of these folders, there are data and gold sub-folders, containing respectively the list of **italian hyponyms** and the list of gold standard **italian hypernyms** in .txt standard. An example of the files format is provided respectively in Listing 1 and Listing 2 to enhance comprehension.

3 Methodology

3.1 Data Reformatting

To evaluate the linguistic skills of Large Language Models (LLMs), the data is converted into JSONL standard. An example of this format is provided in Listing 3 to enhance comprehension. A list of strings representing the possible **choices** available to the model was added. This list is shuffled to create randomness and the position of the true hypernym in the list is indicated by the integer in the **label** key.

4 Find Distractors

Since many models used in computational linguistics are mainly trained using an English vocabulary, it was chosen as a **baseline** for finding distractors

the **Word2Vec** model built directly from the Italian vocabulary provided in the task. To ensure *consistency* in the results, all non-alphabetic characters and non-italian language words were removed using Python functions and **Spacy FastLang** library. The obtained distractors were also filtered to avoid *ambiguous* or *duplicated* distractors.

4.1 Prompts Formulation

The primary objective of this work is to formulate prompts capable to improve the performance and manage the behavior of LLMs. In Listing 4, the list of suitable prompts is provided, saved in JSONL standard.

4.2 Models

To assess the quality of the prompts, the **RoBERTa Fine-Tuned Race** model [3] is used to perform the *multiple choices task*, while **RoBERTa Large IT Multilingual** classifier [2] is used to perform a *Zero-Shot text classification*.

4.3 Prompts Evaluation

The Train Set **confusion matrices** in Figure 1 e Figure 2 graphically summarize the overall performance of the two selected models. (**Note:** Test Set confusion matrices and other statistics can be observed in the Python script.)

5 Final Considerations

The results obtained in Table 1 highlight **challenges** in the model's ability to distinguish hypernyms with high confidence score from distractors.

However, **RoBERTa Fine-Tuned Race** model seems to work better than the classifier, especially in **Prompt 1**. By employing *fine-tuning*, the overall results could potentially increase. Future improvements could also involve the utilization of an *external Italian vocabulary* to enhance the quality of distractors.

6 Code Instructions

The Python script was run on Google Colab and the dataset was uploaded on Google Drive to make it accessible in the notebook. GPU is necessary to run models and avoid execution time problems

7 Tables and Figures

Listing 1: Example of one sample of Training Set data file (each line corresponds to a term and its type).

sesto	Concept
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Listing 2: Example of one sample of Training Set gold file (each line corresponds to the same line in the corresponding data file).

grado	numero	ordinale
frazione		carica

Listing 3: Example of one sample of Train Set reformatting

```
{ 'id': 0, 'text': 'sesto ',
  'choices': [ 'inalterabilita ',
               'grado ', 'bareggio ',
               'arteriosclerotico ' ], 'label': 1 }

{ 'id': 1, 'text': 'sesto ',
  'choices': [ 'lateralizzazione ',
               'accordatura ', 'provvedimento ',
               'numero ordinale ' ], 'label': 3 }

{ 'id': 2, 'text': 'sesto ',
  'choices': [ 'frazione ',
               'limitato ', 'pellicola ',
               'concorrenza ' ], 'label': 0 }

{ 'id': 3, 'text': 'sesto ',
  'choices': [ 'capodoglio ',
               'carica ', 'asciugacapelli ',
               'ridisegnare ' ], 'label': 1 }
```

Listing 4: Prompts list

```
{"prompt": "Il termine '{text}' pu\u00f2
```

```
essere iperonimo di: \n
```

```
a) {option1} \n
b) {option2} \n
c) {option3} \n
d) {option4}"}
```

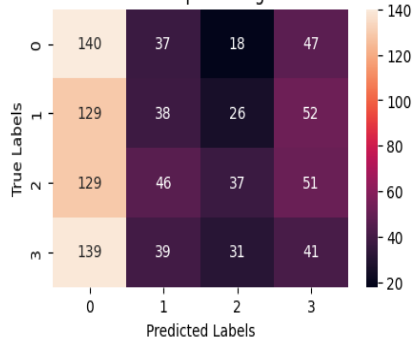
```
{"prompt": "Dato il termine '{text}',
quale tra le seguenti parole \u00e8
un suo iperonimo? \n
```

```
a) {option1} \n
b) {option2} \n
c) {option3} \n
d) {option4}"}
```

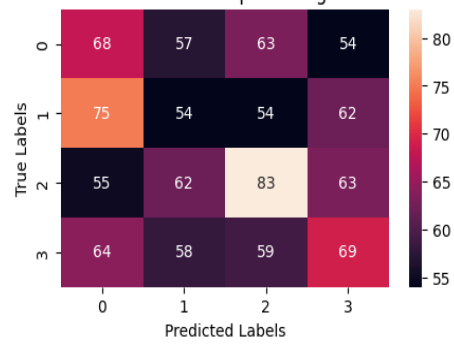
```
{"prompt": "Scegli l'iperonimo
del termine '{text}': \n
```

```
a) {option1} \n
b) {option2} \n
c) {option3} \n
d) {option4}"}
```

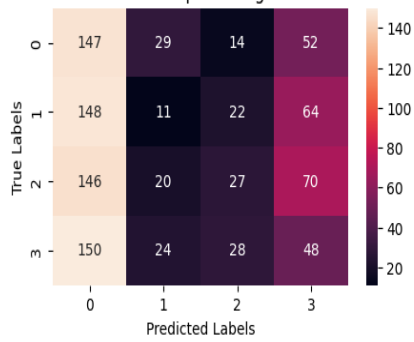
Train Confusion Matrix for Prompt 1 using Roberta Fine-Tuned Race



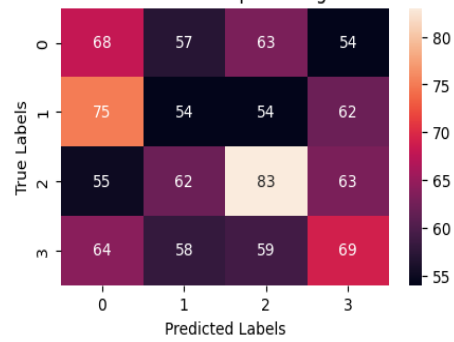
Train Confusion Matrix for Prompt 1 using Roberta Classifier



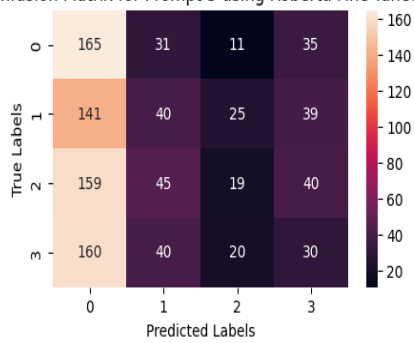
Train Confusion Matrix for Prompt 2 using Roberta Fine-Tuned Race



Train Confusion Matrix for Prompt 2 using Roberta Classifier



Train Confusion Matrix for Prompt 3 using Roberta Fine-Tuned Race



Train Confusion Matrix for Prompt 3 using Roberta Classifier



Figure 1: Train Confusion Matrices using Roberta Fine-Tuned Race

Figure 2: Train Confusion Matrices using Roberta Large IT Multilingual

	Prompt 1	Prompt 2	Prompt 3
Train Accuracy Roberta Fine-Tuned Race	0.256	0.233	0.254
Train F1-score Roberta Fine-Tuned Race	0.231	0.192	0.209
Test Accuracy Roberta Fine-Tuned Race	0.300	0.283	0.277
Test F1-score Roberta Fine-Tuned Race	0.273	0.247	0.234
Train Accuracy Roberta Large IT Multilingual	0.274	0.274	0.274
Train F1-score Roberta Large IT Multilingual	0.274	0.274	0.274
Test Accuracy Roberta Large IT Multilingual	0.247	0.247	0.247
Test F1-score Roberta Large IT Multilingual	0.246	0.246	0.246

Table 1: Performance Comparison Table

References

- [1] Delli Bovi Claudio Espinosa-Anke Luis Oramas Sergio Pasini Tommaso Santus Enrico Shwartz Vered Navigli Roberto Camacho-Collados, Jose and Horacio, Saggion. *SemEval-2018 Task 9: Hypernym Discovery, Proceedings of the 12th International Workshop on Semantic Evaluation*. 2018.
- [2] Jiva. xlm-roberta-large-it-mnli. [Hugging Face](#).
- [3] LIAMF-USP. roberta-large-finetuned-race. [Hugging Face](#).